PHYS 7501:
Particles and Nuclei I

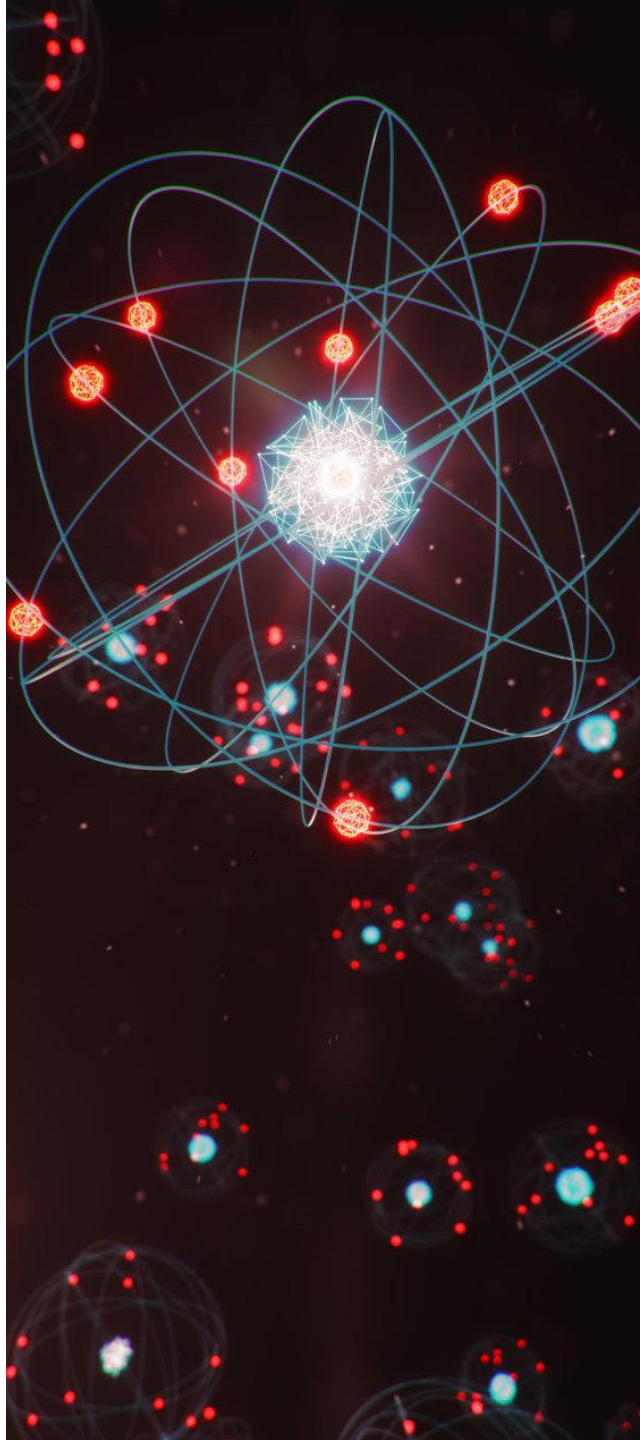
Tues./Thurs. 2 – 3:20PM
Clippinger 105

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Available via e-mail & Teams



UPDATED SCHEDULE

Date	Chapters	Topic	Due
August 26	1,2	Nuclear Properties	
August 28	1,2	Nuclear Properties	
September 2	1,2,5	Nuclear Phenomenology	
September 4	1,2,5	Nuclear Phenomenology	
September 9	6	Nuclear Structure	HW #1
September 11	6	Nuclear Structure	
September 16	6	Nuclear Structure	
September 18	3	Radioactive Decay	
September 23	7	Alpha Decay	HW #2
September 25	8	Beta Decay	
September 30	9	Gamma Decay	
October 2	11	Fission	
October 7	**Midterm Exam**		
October 9	5,6	Nuclear Forces	HW #3
October 14	10	Nuclear Reactions	
October 16	10	Scattering	
October 21	10	Direct Reactions	
October 23	10	Resonant Reactions	
October 28	10	Statistical Reactions	HW #4
October 30	10	Other Reaction Types	
November 4	10,12	Nuclear Astrophysics	
November 6	12	Nuclear Astrophysics	
November 11	Veteran's Day – No Class		
November 13	12	Nuclear Astrophysics	HW #5
November 18	13,17,18	Experimental Techniques	
November 20	4	Special Topics: Nuclear Medicine	
November 25	15	Special Topics: Nuclear Energy	
November 27	Thanksgiving Break – No Class		
December 2		Special Topics: TBD	HW#6
December 4		**Final Presentations Group 1**	
December 9	**Final Exam Presentation Group 2: 12:20PM**		

NUCLEAR PROPERTIES

Modern Nuclear Chemistry, Chapter 2

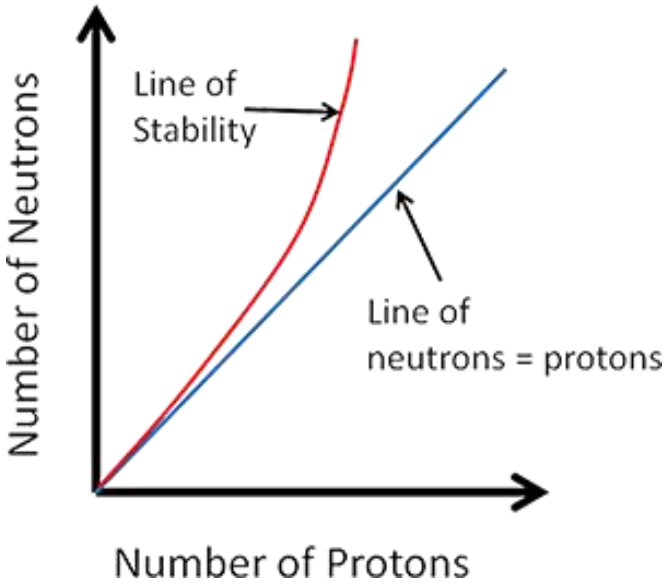
Isomer comes from “equal parts” because it’s the same nucleus but a different state

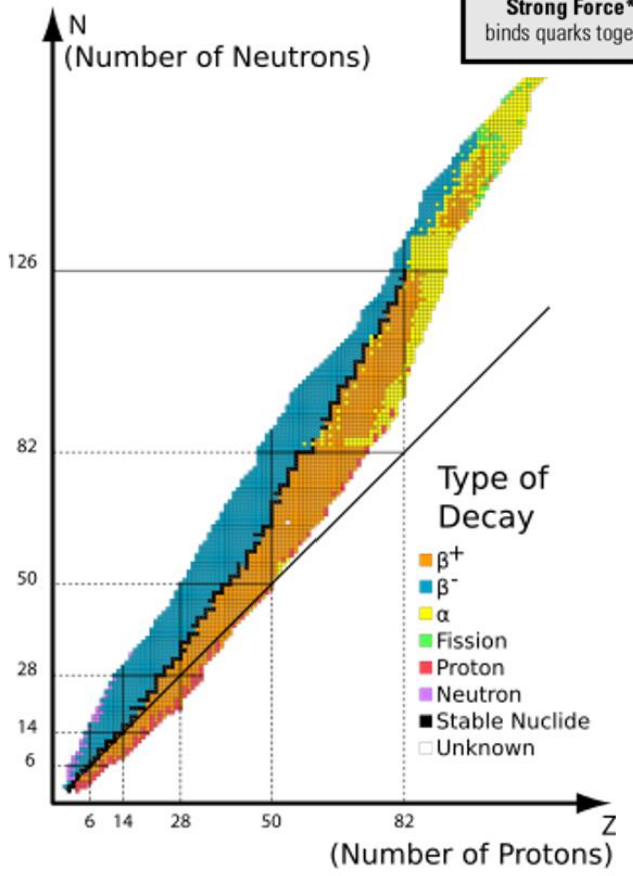
Nuclear Forces and Stability

- **Stable Nuclei** do not decay via the Strong Force – they may transform via β or α emission with long lifetimes
- Characteristics of Nuclides
  - Light nuclides have N = Z
  - Heavy nuclides have more neutrons N > Z

Fundamental Force Particles

Force	Particles Experiencing	Force Carrier Particle	Range	Relative Strength*
Gravity acts between objects with mass	all particles with mass	graviton (not yet observed)	infinity	much weaker ↓ much stronger
Weak Force governs particle decay	quarks and leptons	W^+, W^-, Z^0 (W and Z)	short range	
Electromagnetism acts between electrically charged particles	electrically charged	γ (photon)	infinity	
Strong Force** binds quarks together	quarks and gluons	*g* (gluon)	short range	





Nuclear Landscape Reminders

- Recall that nuclides are specified by
  - **Z**: Atomic Number = number of protons
  - **N**: Neutron Number = number of neutrons
  - **A**: Mass Number = number of nucleons = **Z + N**

To get familiar with the Nuclear Chart, work together to determine the following:

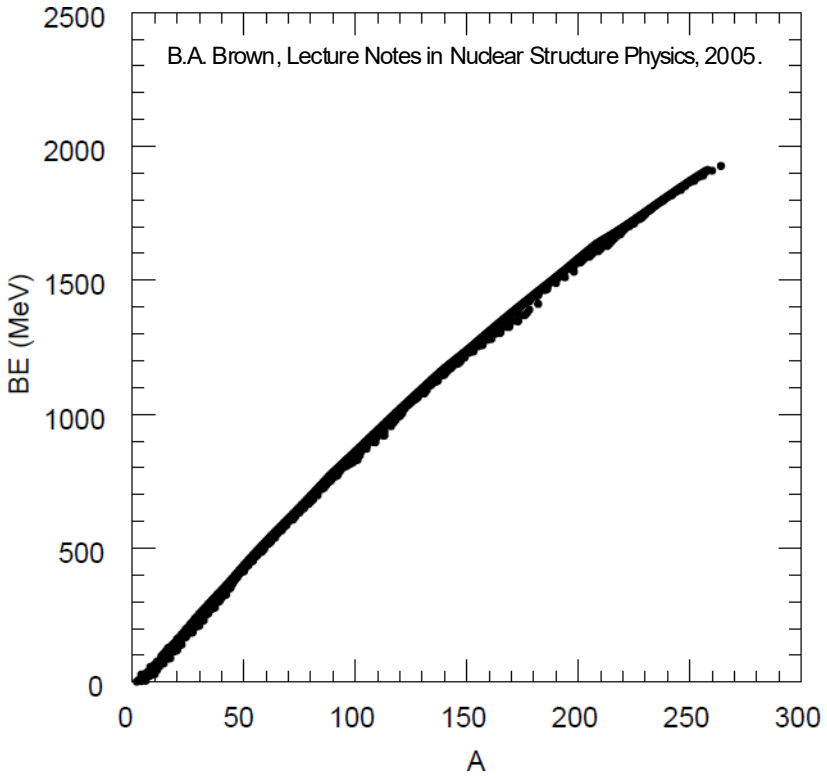
1. Number of protons in ^{48}Ca: Z = 20
2. Number of neutrons in ^{70}Cu: Z = 29, N = A – Z = 70 – 29 = 41
3. Is this reaction possible?
   a) $^{12}C(\alpha,\gamma)^{16}O$ Yes, ^{12}C + ^{4}He gives us Z=8 and N=8 which is ^{16}O
   b) $^{126}Te(n,\gamma)^{127}I$ No, ^{126}Te + n gives us ^{127}Te

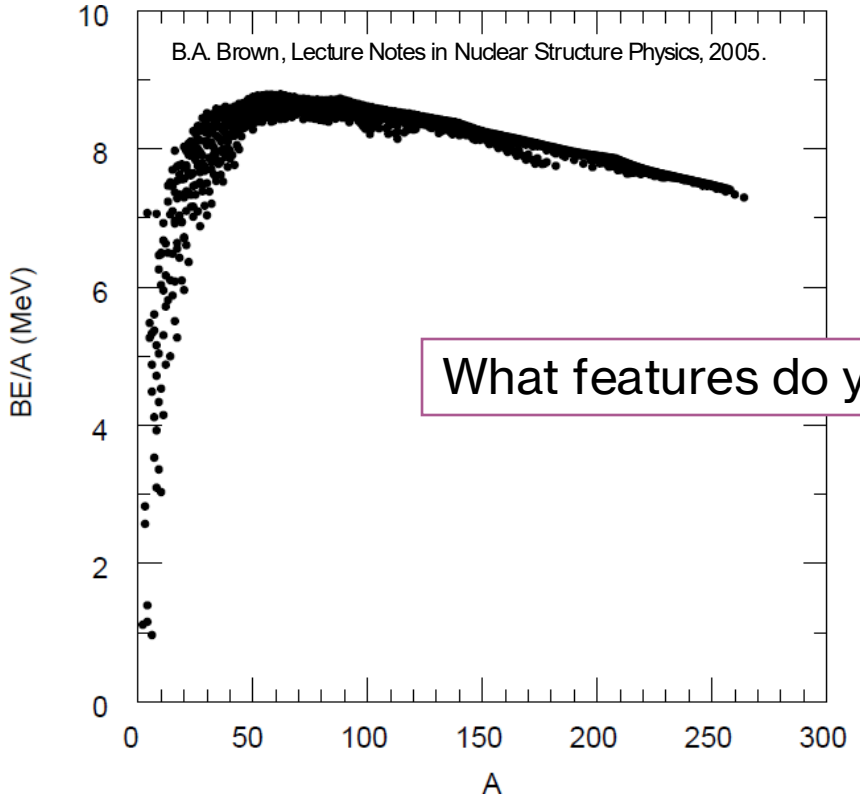
Binding Energy

- Masses are thought of to be *almost* integer values being multiplied by the mass of the Hydrogen atom
  - However, precision mass measurements show deviations from this description called “mass defects”
- Nucleons within a nucleus are bound together and that binding requires energy
- The **binding energy** (BE) is energy required to split a nucleus into its constituents
- Mass of nucleus: $\boldsymbol{m(Z,N) = Zm_p + Nm_n - BE}$
- Binding Energy is important because it describes
  - forces between nucleons
  - stability of nucleus
  - energy released or required in nuclear decays or reaction
- Relies on precise measurements of nuclear mass to be calculated
- Essentially reduces the overall mass of the system

Binding Energy

- BE increases linearly with A
- It's not surprising that more nucleons would lead to more binding since nucleons are prone to attraction via the Strong Force
- The more interesting way to visualize things is in terms of BE/A or "binding energy per nucleon"

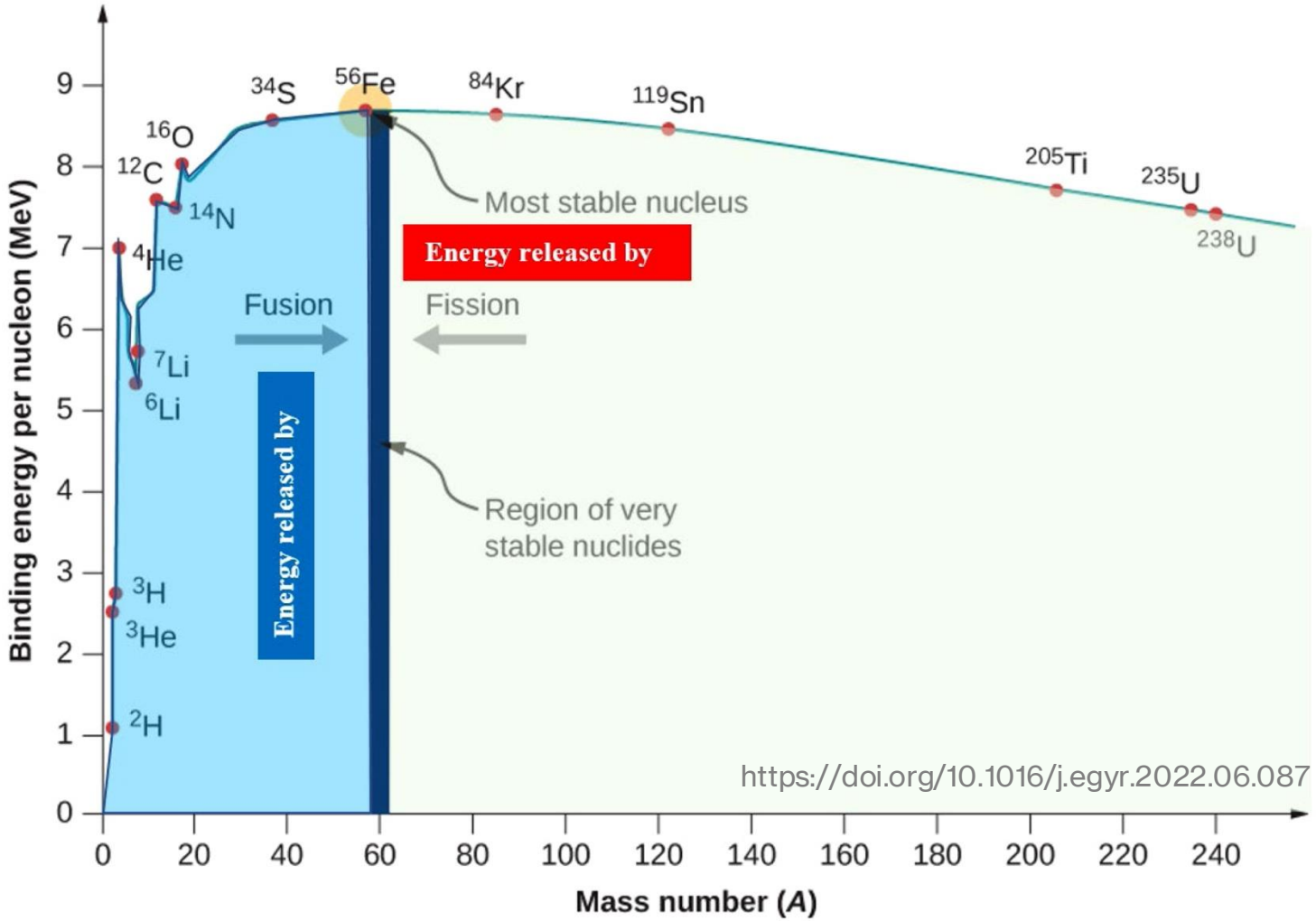




What features do you notice?

Binding Energy

- Average BE/A is 8 MeV per nucleon (MeV/u or MeV/A)
- ^{56}Fe is the most bound nucleus
  - This is why nuclear fusion in star cores ends with iron!
- At low A, attractive forces between nucleons dominate over repulsive electrostatic forces.
- At high A, repulsive electrostatic forces begin to dominate, and these forces tend to break apart the nucleus rather than hold it together.



Give it a try

$$m(Z,N)c^2 = Zm_p + Nm_n - BE$$

Calculate the binding energy per nucleon for ^{4}He (alpha particle)

$m_p = 1.007825u$
$m_n = 1.008665u$
$1u = 931.5\ \text{MeV}/c^2$

Solution:

$$BE = (Zm_p + Nm_n - m(Z,N))c^2$$
$$BE = (2m_p + 2m_n - m(^4He))c^2$$

Look up m(^{4}He) in AME database (4.00260325413u)

$$BE = (2m_p + 2m_n - m(^4He))c^2$$
$$BE = (0.030378u)c^2$$
$$BE = (0.030378)(931.5\frac{MeV}{c^2})c^2$$
$$BE = 28.3\ MeV$$

$$\frac{BE}{A} = 7.074\ MeV/nucleon$$

Compare to previous slide and AME Database

Nuclear Masses and Energy

- Nuclear Mass: $\boldsymbol{m(Z,N)c^2 = Zm_p + Nm_n - BE}$
- Atomic mass: $\boldsymbol{\mathrm{M}(Z,N)c^2 = Z(m_p + m_e) + Nm_n - BE}$

- It's often easier to use ***mass excess*** for calculations
  - Usually denoted as Δ or ME
  - The difference between atomic mass and A x u: $\boldsymbol{\Delta M(A,Z) = M(Z,N) - A \times u}$
  - Defined so that ^{12}C has a mass excess of zero!
  - Values can be looked up on AME

  - **Example**: Mass excess of ^{64}Cu
    - $\boldsymbol{\Delta M(^{64}Cu) = 63.929764u - 64 \times u = -65.42\ \mathrm{MeV}}$ (recall that $1u$ = 931.5 MeV/c^2)
- For most calculations, we will be using mass excess from now on!

Q-values

Recall our transmutations:
Reactions: A + b → c + D
Decay: A → B + particle

- Energy change in nuclear reactions is called the *Q-value*

$$Q = \sum_{reactants} \Delta(A,Z) - \sum_{products} \Delta(A,Z)$$

^{68}Se + p → ^{69}Br + γ
reactants *products*

- For example: ^{68}Se(p,γ)^{69}Br

- $Q_{^{68}Se(p,\gamma)^{69}Br} = \Delta(^{68}Se) + \Delta(p) - \Delta(^{69}Br)$

$$= (-54.189\ MeV) + (7.288\ MeV) - (-46.260\ MeV)$$

$$= -0.641\ \text{MeV}$$

Now check it with the Q-Calc on the NNDC

^{68}Se + p

Reaction Products	Q-Value (keV)	
^{69}Br + γ	-6.4E+2	*4*

Group Work: Q-Values

- Calculate the following Q-values (then check with Q-Calc)

1. $^{12}C(\alpha,\gamma)^{16}O$

$$Q_{^{12}C(\alpha,\gamma)^{16}O} = \Delta(^{12}C) + \Delta(^{4}He) - \Delta(^{16}O) = 0 + 2.4249 - (-4.737) = 7.1622 \text{ MeV}$$

2. $^{100}Mo(n,\gamma)^{101}Mo$

$$Q_{^{100}Mo(n,\gamma)^{101}Mo} = \Delta(^{100}Mo) + \Delta(n) - \Delta(^{101}Mo) = -86.193 + 8.0713 - (-83.520) = 5.3983 \text{ MeV}$$

3. $^{70}Cu\ \beta^-$ decay

$$Q_{^{70}Cu\ \beta-} = \Delta(^{70}Cu) - (\Delta(^{70}Zn) = -62.976 - (-69.564) = 6.588 \text{ MeV}$$

4. $^{238}U\ \alpha$ decay

$$Q_{^{238}U} = \Delta(^{238}U) - (\Delta(^{4}He) + \Delta(^{234}Th)) = 47.3077 - (2.4249 + 40.613) = 4.269 \text{ MeV}$$

Separation Energies

- **Separation Energy** of a nucleon is the energy required to remove a single nucleon from a nucleus.
- One Proton Separation Energy:

$$S_p(Z,N) = \Delta(Z-1,N) + \Delta(p) - \Delta(Z,N)$$

- Two Proton Separation Energy:

$$S_{2p}(Z,N) = \Delta(Z-2,N) + 2 \times \Delta(p) - \Delta(Z,N)$$

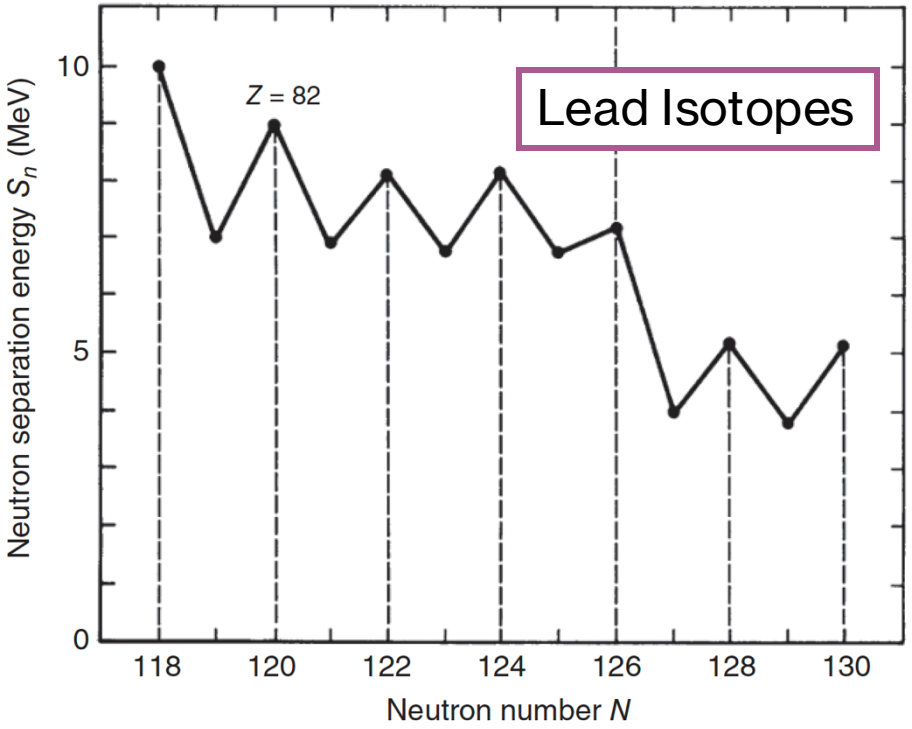
- One Neutron Separation Energy:

$$S_n(Z,N) = \Delta(Z,N-1) + \Delta(n) - \Delta(Z,N)$$

- Two Neutron Separation Energy:

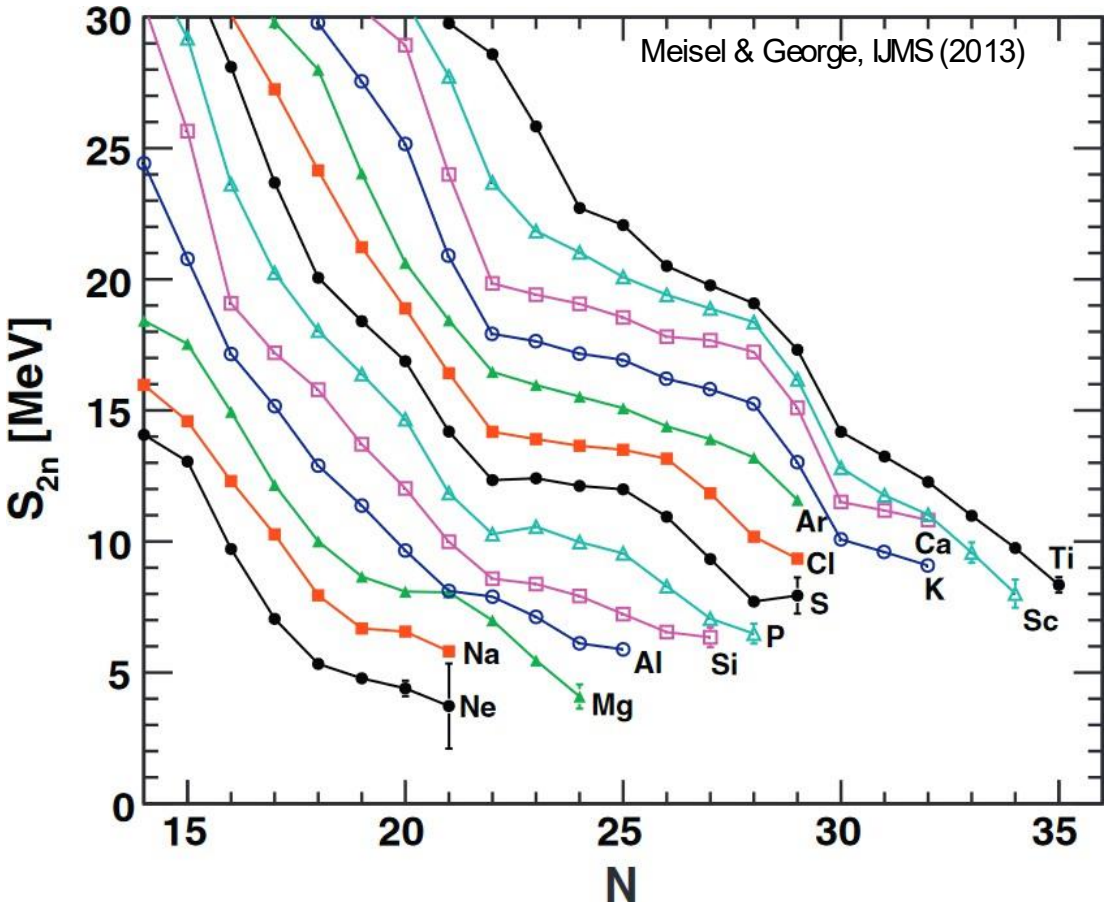
$$S_{2n}(Z,N) = \Delta(Z,N-2) + 2 \times \Delta(n) - \Delta(Z,N)$$

Trends in Separation Energies



Lead Isotopes

Notice anything interesting?



Let's do more math ☺

Calculate the One and Two Neutron Separation Energies for

^{45}Sc

Z=21
N=24

$$S_n(Z,N) = \Delta(Z,N-1) + \Delta(n) - \Delta(Z,N)$$
$$= \Delta(21{,}23) + \Delta(n) - \Delta(21{,}24) = -37.816 + 8.071 - (-41.072) = 11.327\ MeV$$

$$S_{2n}(Z,N) = \Delta(Z,N-2) + 2\ \times\ \Delta(n) - \Delta(Z,N)$$
$$= \Delta(21{,}22) + 2\Delta(n) - \Delta(21{,}24) = -36.1881 + 2 \times 8.071 - (-41.072) = 21.0259\ MeV$$

^{235}U

Z=92
N=143

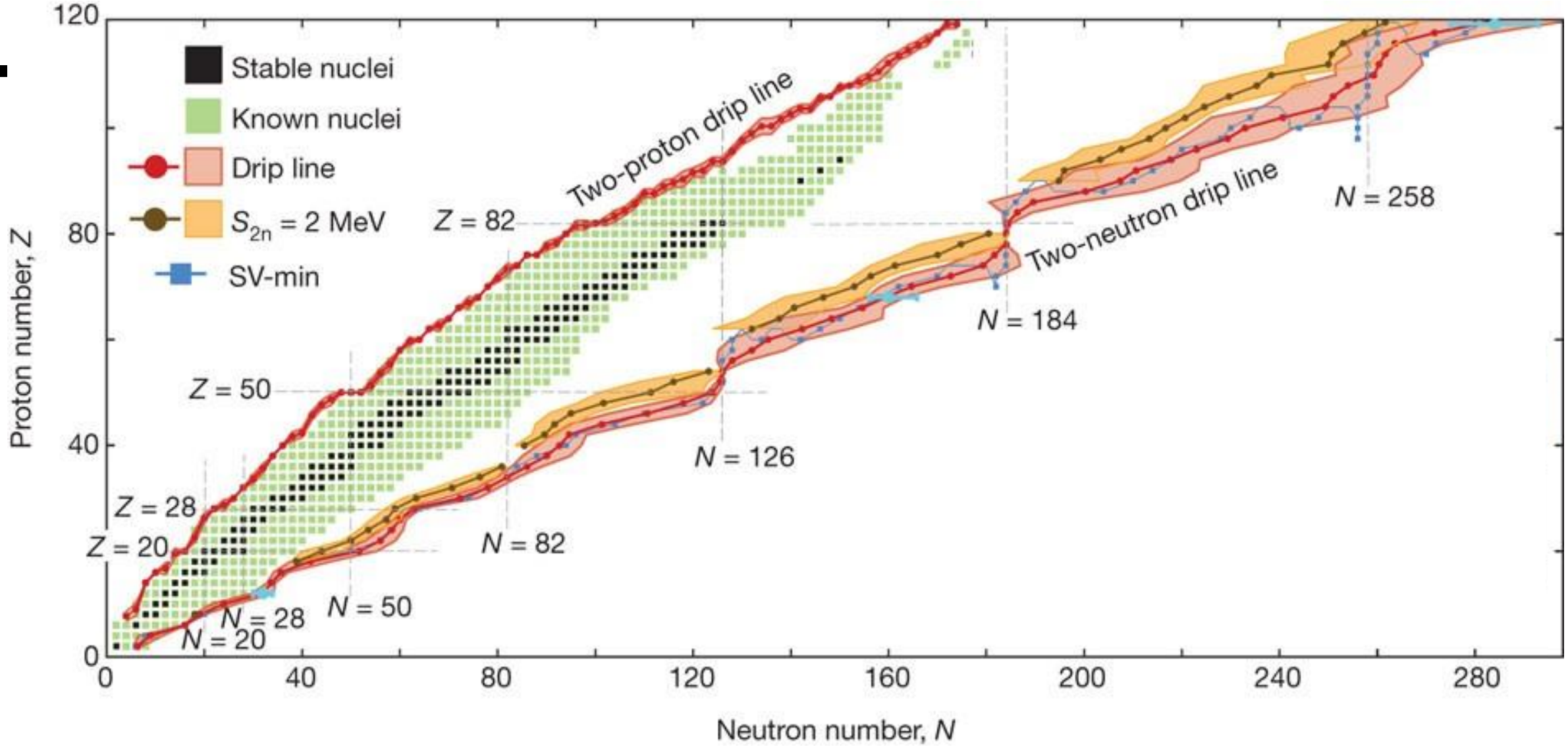
$$S_n(92{,}143) = 5.2975\ MeV$$
$$S_{2n}(92{,}143) = 12.143\ MeV$$

^{60}Zn

Z=30
N=30

$$S_n(30{,}30) = 15.030\ MeV$$
$$S_{2n}(30{,}30) = 28.02\ MeV$$

Masses Define the Nuclear Chart



J Erler *et al. Nature* **486**, 509-512 (2012) doi:10.1038/nature11188

Semi-Empirical Mass Formula

- C. von Weizsäcker developed the first theory for nuclear masses (Z. Phys. A 1935), now known as the semi-empirical mass formula (SEMF)
  - Also known as the Weizsäcker formula, Bethe–Weizsäcker formula, or Bethe–Weizsäcker mass formula
  - This is a parameterization of the total binding energy of the nucleus
- The **Liquid Drop Model**: Assume nuclei behave as if they are incompressible uniformly charged liquid drops and approximate the nucleus as a sphere with a uniform interior density

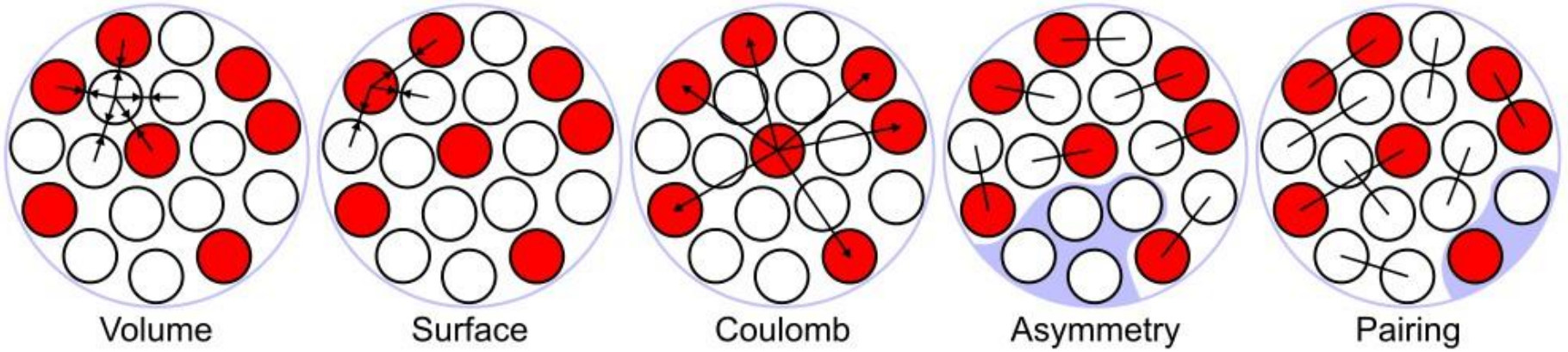


Liquid Drop:

- Short-range intermolecular forces.
- Density independent of drop size.
- Heat required to evaporate fixed mass independent of drop size.

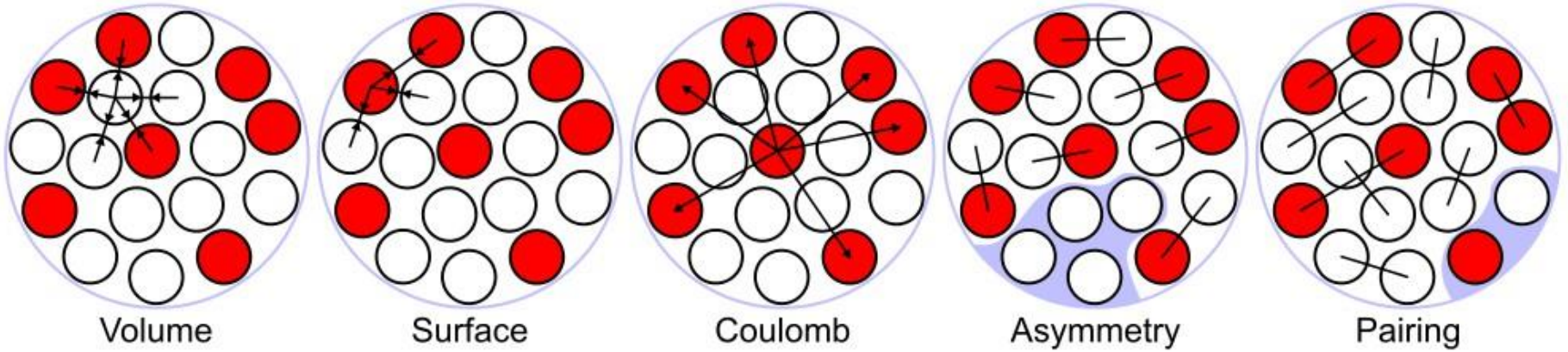
Nucleus

- Nuclear force short range.
- Density independent of nuclear size.
- B/A ~ constant



- We want to more carefully consider all interactions in a nucleus
- All nucleons are attracting each other via the Strong Force, generating some overall binding energy
- However, nucleons at the surface don't have a neighbor, reducing binding energy
- Protons repel each other due to the Coulomb force, reducing binding energy
- p-n interaction is stronger than p-p or n-n, favors N=Z but we know this only happens for a few nuclei, reducing binding energy
- Nucleons prefer to be paired, so there is an enhancement for even-even nuclei, reduction for odd-odd nuclei, and neutral for even-odd nuclei

START HERE AS REMINDER



$$BE(A,Z) = Volume - Surface - Coulomb - Asymmetry \pm Pairing$$

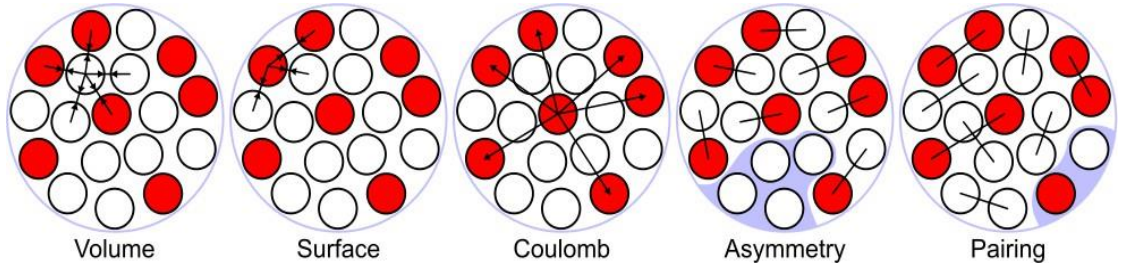
$$BE(A,Z) = a_v A - a_s A^{\frac{2}{3}} - a_c \frac{Z^2}{A^{\frac{1}{3}}} - a_a \frac{(A-2Z)^2}{A} \pm \delta$$

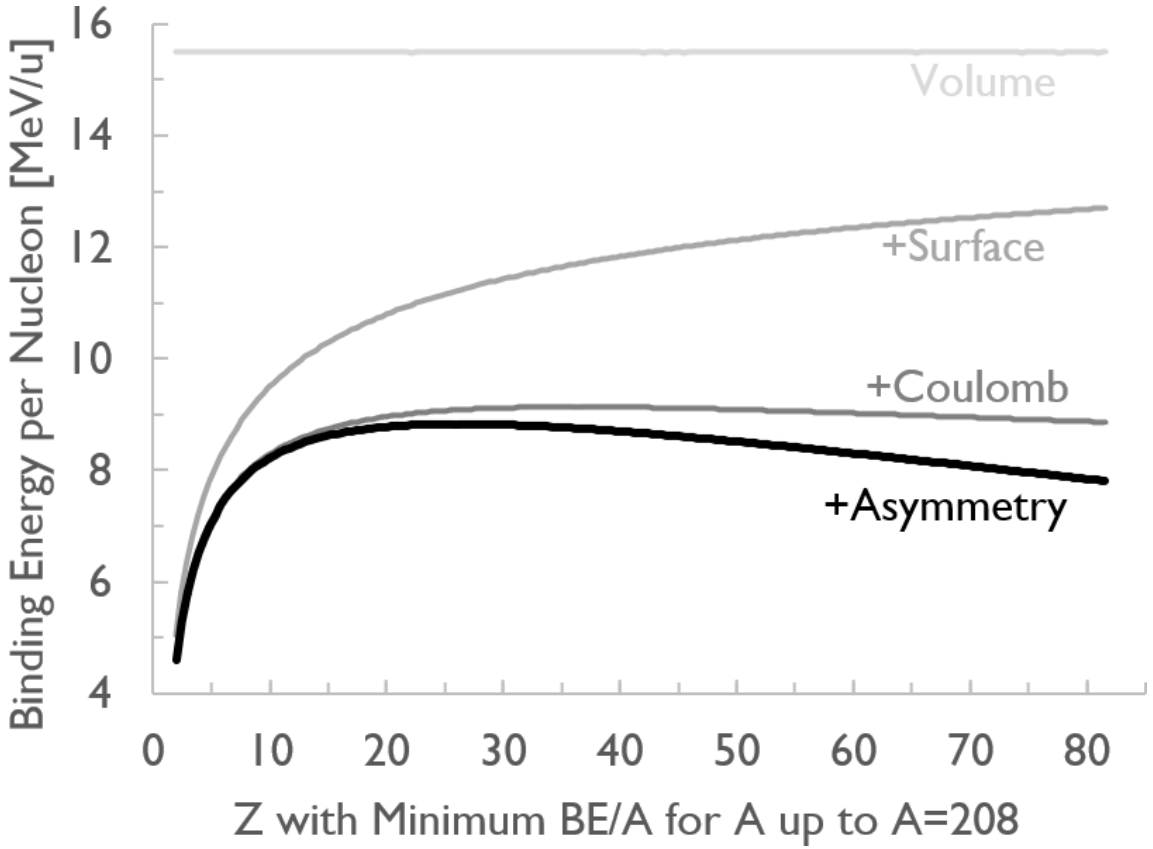
There are other forms of this equation, this is the book's notation

Volume (a_v)
Surface (a_s)
Coulomb (a_c)
Asymmetry (a_a)
Pairing (δ):
- Even-Z, Even-N: +δ
- Odd, Odd-N: -δ
- Even-Z, Odd-N: 0

a_i values are fit to empirical mass data

SEMF





Volume (a_v): $a_v = 15.56\ MeV$

Surface (a_s): $a_s = 17.23\ MeV$

Coulomb (a_c): $a_c = 0.7\ MeV$

Asymmetry (a_a): $a_a = 23.285\ MeV$

Pairing (δ): $\delta = \frac{11}{A^{1/2}}\ MeV$

- Even-Z, Even-N: +δ
- Odd, Odd-N: -δ
- Even-Z, Odd-N: 0

What's Next?

- Still more to do in Chapter 2
- Shapes, Sizes of Nuclei
- HW will be sent around in the next few days